

# Parvez Akhtar

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## PROFESSIONAL SUMMARY

Senior Generative AI Engineer with 5+ years building high-scale distributed systems and production-grade AI agents. Leads the design and deployment of agentic solutions spanning multi-agent orchestration (LangGraph, MCP), RAG (Retrieval-Augmented Generation) over vector search, LLM fine-tuning and evaluation, prompt engineering, and agent observability and guardrails. Reduced inference costs by ~90% for production extraction engines while taking experimental GenAI to reliable, cost-efficient enterprise scale.

## TECHNICAL SKILLS

**LLMs & Generative AI:** Transformer Architecture & Attention, Fine-tuning (LoRA/QLoRA, Hugging Face Transformers), Hyperparameter Tuning, Prompt Engineering, Structured Outputs, Embeddings & Vectorization, Pre-trained & Deployed Models.

**AI Agents & Orchestration:** Multi-Agent Orchestration (LangGraph, OpenAI Agents SDK, Anthropic SDK, LangChain), Tool-Use & Function Calling, Multi-Step Planning, Agent Memory (Short-term & Long-term), Human-in-the-Loop, MCP (Model Context Protocol), Swarm-based Orchestration.

**RAG & Vector Search:** RAG Pipelines over Documents/Databases/APIs, Hybrid Search (SPLADE + Dense Retrieval), Cross-Encoder Re-ranking, Reciprocal Rank Fusion, Vector Databases (Qdrant, PGVector), Vector Indexing, LlamaIndex.

**Evals & Observability:** LLM Tracing & Eval (Langfuse, RAGAS, OpenTelemetry), Guardrails & Prompt-Injection Defense, CI/CD Evaluation Gates, SigNoz, CloudWatch, Prometheus.

**ML & Data:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Data Preprocessing, Large-Scale Datasets, Snowflake Cortex, Apache Airflow, Kafka, Elasticsearch, ETL Pipelines, MongoDB, Redis, PostgreSQL.

**Languages & Cloud:** Python, Java, TypeScript, SQL; AWS (SageMaker, Bedrock, Lambda, ECS), Azure OpenAI, GCP, Kubernetes, Docker, Terraform, Temporal.io.

## PROFESSIONAL EXPERIENCE

### Bigstep Technologies

Remote

*Senior AI/ML Engineer, Lead Architect*

04/2025 – Present

- As **Lead Architect**, led the design and deployment of cloud-native GenAI features across client accounts spanning high-growth startups (Seed to Series F) and global enterprises including **CNBC London** and **VTS**; ran technical design sessions that translated ambiguous business goals into scalable, secure production AI architectures.
- **AI Agents & Orchestration:** Built a **serverless multi-agent** framework for Salesforce automation with **tool/function calling**, **multi-step planning**, and agent memory over **MCP (Model Context Protocol)** and **swarm-based orchestration**. Cut idle compute via a pure **pay-per-execution** model (saving ~\$45k/year).
- **RAG & Vector Search:** Built a **Hybrid Search** pipeline fusing **SPLADE** (learned sparse) and **Sentence-BERT** (dense vector) retrieval via **Reciprocal Rank Fusion**, then layered **Cross-Encoder re-ranking** over the top candidates, improving **NDCG@5** by ~15 points (**0.71 to 0.86**).
- **Client, Outcomelabs.ai (GenAI Platform):** Architected a **Field-Aware Extraction Engine** processing **2M+ records/month**. Built a **"Strategy Router"** that classifies each field by complexity and routes ~60% to deterministic parsers, reserving the LLM for the rest, cutting token spend ~40% with no measurable accuracy loss.
- **Client, VTS (current), Lease Abstraction:** Built an agentic **document-intelligence** pipeline that abstracts structured terms from commercial real-estate leases with **structured-output validation**. Diagnosed a **10-point accuracy regression** and redesigned **prompt strategy** and image preprocessing to recover the full drop, restoring field-level extraction to its ~88% **baseline** and unblocking downstream automation.
- **Scale & Impact:** Scaled the platform from MVP to production, underpinning a **\$20M+ enterprise deal pipeline** within 6 months while continuously optimizing models for performance and cost efficiency.
- **Large-Scale ML:** Constructed an entity-matching pipeline on 22M records using an ensemble of **TF-IDF/Cosine Similarity** and 7 text metrics feeding an **XGBoost classifier** (85.3% AUC).

### Kuttl (Incubated at IIM Lucknow)

Remote

*Founding Engineer*

12/2023 – 03/2025

- Joined as **Founding Engineer** for an IIM Lucknow-incubated, GenAI-native orchestration platform; built the MVP from 0-to-1 that secured a term sheet, then documented and standardized the codebase for a **zero-downtime handover** to the incoming engineering team.
- **Fine-tuned Llama-3-8B** (using LoRA) to replace GPT-4 for structured entity extraction, achieving **98% schema compliance** while reducing per-token inference costs by **90%** and latency by 300ms.

- **Evals:** Operationalized the ML lifecycle with **CI/CD evaluation gates (RAGAS)** scoring **faithfulness and answer-relevancy**, automatically blocking deployments below threshold and catching quality regressions before production.
- Applied **semantic caching** and request batching for **RAG** endpoints, improving **P99 latency by 40%** during high-concurrency bursts.
- Engineered a distributed document-parsing pipeline using **FastAPI** and **Temporal.io**, processing 50k+ daily messages with resilient retry logic and dead-letter queues to ensure no data loss.

#### **Amplicomm**

Bengaluru, India

*SDE-2*

05/2023 – 12/2023

- Led a team of 3 through a **Strangler-Fig migration** of a legacy monolith to Spring Boot microservices, enabling independent, incremental deploys with instant rollback; enforced **SonarQube** quality gates that cut cyclomatic complexity ~42%.
- Established a centralized identity broker using **Keycloak** and **Spring Security**, replacing stateful DB lookups with **stateless JWT** validation to cut login latency by 30%.
- Engineered a **Schema-per-Tenant** architecture with **RBAC (Role-Based Access Control)** for 21 enterprise clients, ensuring data integrity via automated checksum reconciliation pipelines.

#### **Amazon**

Bengaluru, India

*Software Development Engineer (SDE)*

03/2021 – 03/2023

- Owned the full SDLC for high-scale microservices, co-developing the SmartPOS billing engine and scaling it to support **3,000+ enterprise merchants** with 99.99% availability.
- Optimized cloud infrastructure costs by **25% (~\$100k/yr)** via strategic re-architecture of ECS services to utilize **Spot Instances** and custom auto-scaling policies.
- Developed a data-driven testing framework and **CloudWatch** service-level alarms, identifying **N+1 query bottlenecks** that cut p95 API latency by **20%** and surfaced performance regressions before release.
- Executed the zero-downtime migration of 6 core microservices from GCP to AWS, ensuring data consistency and seamless customer transition.

#### **Publicis Sapient**

Remote, India

*Associate Data Engineer*

09/2020 – 03/2021

- Designed high-frequency data ingestion pipelines using **Apache Airflow** and **Crawlera** to scrape pricing data from Amazon, Walmart, and Target, bypassing complex anti-bot protections for competitive intelligence.
- Refined **Elasticsearch** indexing strategies and aggregation queries, reducing query latency by ~40% to enable sub-second dashboard load times for brand performance analytics.

### EDUCATION

**National Institute of Technology Delhi**  
*B.Tech in Computer Science & Engineering*

Delhi, India  
2016 – 2020

### ACHIEVEMENTS

**Amazon "On the Hero Road" Award:** Recognized for exceptional impact on team delivery and engineering excellence.

**Udacity Computer Vision Nanodegree:** Full scholarship recipient (sponsored by Facebook).

**Deep Learning Specialization:** Completed 5-course specialization (Coursera) with perfect score.